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RESEARCH ARTICLE

Fine-Tuned YOLO Model for Monitoring Children Across Medical Scenes Based on a Large-Scale Real-World Dataset for Children Detection

SAMUEL DIOP[1,2], FRANÇOIS JOUEN[1,2], JEAN BERGOUNIOUX[3], AND IMEN TRABELSI[1,2]
[1]CHArt, EPHE, Paris 8, UPEC, CY, Université PSL, 75014 Paris, France
[2]Systèmes Intelligents en Neurologie et Réanimation Pédiatrique (R2P2), 92380 Garches, France
[3]UVSQ, 2IC UMR 1173, Hôpital Raymond Poincaré (AP-HP), Service de Réanimation et Neurologie Pédiatriques, Université Paris-Saclay, 92380 Garches, France

Corresponding authors: Samuel Diop (samuel.diop@etu.ephe.fr) and Imen Trabelsi (imen.trabelsi@ephe.psl.eu)

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ABSTRACT Accurate detection and monitoring of children in medical settings using computer vision systems present unique challenges due to anatomical differences, environmental complexity, and stringent privacy constraints. This paper introduces YOLOCDD, a fine-tuned YOLOv11-based model optimized for child detection in medical scenes, supported by the Child Detection Dataset (CDD)—a large-scale, real-world dataset comprising 1,928 annotated images of children across diverse age groups and interaction scenarios. Unlike existing datasets that rely heavily on synthetic data or controlled environments, CDD captures realistic medical and everyday settings, including occlusions, multi-child interactions, and dynamic lighting conditions. Our model achieves a mean average precision (mAP@50) of 0.953 in medical environments, significantly outperforming general-purpose detectors like YOLOv11x (mAP@50: 0.606).This work bridges critical gaps in pediatric medical AI by providing a scalable, privacy-compliant dataset, delivering a high-precision detection model, and showcasing clinical applicability in neurological diagnostics. The dataset is publicly available to foster further research in child-centric computer vision.

INDEX TERMS Child detection, medical monitoring, computer vision, YOLO, deep learning, video-EEG, pediatric assessment.

I. INTRODUCTION

The accurate detection and tracking of children through computer vision systems represents a critical advancement in medical monitoring and assessment, particularly in neurological diagnostics [1], [2], where continuous observation is essential [3]. Current clinical practice in pediatric neurology relies heavily on manual observation, creating significant burdens on healthcare providers while potentially missing subtle diagnostic indicators. Automation of this process, especially in video-EEG monitoring, could improve early detection and treatment monitoring of conditions such as infantile spasms, providing physicians with objective, quantifiable data while reducing the resource intensity of continuous observation [4].

However, the development of robust child detection systems presents unique challenges that extend beyond traditional computer vision problems. These challenges stem from fundamental anatomical differences between children and adults, most notably in body proportions and movement patterns. As shown in Fig. 1, children exhibit markedly different morphological characteristics, with a head-to-body

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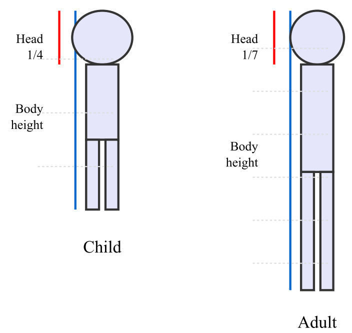


FIGURE 1. **Key Anatomical Differences. Head is 1/7 of body in adult vs 1/4 in infant. Proportionally longer trunk in infant. Different limb proportions. Less developed neck muscles in infant.**

ratio of approximately 1/4 compared to 1/7 in adults, along with proportionally longer trunks and less developed neck musculature [5]. These distinct physical characteristics render standard person detection models, typically optimized for adult proportions, inadequate for reliable child detection and tracking [6].

The complexity of medical environments further compounds these challenges. The presence of medical equipment, particularly EEG helmets and monitoring devices, frequently alters or obscures the child's appearance. Dynamic interactions between patients, medical staff, and caregivers create complex scenes with multiple participants and varying degrees of occlusion [7]. While general object detection has benefited from extensive datasets like COCO [8], with its 118,000 images, the largest available child detection dataset, SyRIP, contains only 1,700 images, of which merely 700 are real-world samples [9]. This limitation becomes particularly critical in medical contexts, where the detection system must be sensitive to subtle movements and posture changes indicative of neurological events.

Moreover, stringent legal and privacy constraints, particularly the General Data Protection Regulation (GDPR) and the Children's Online Privacy Protection Act (COPPA) [10], create significant barriers to data set collection and sharing, forcing researchers to rely heavily on synthetic data that often do not capture the nuances of real-world scenarios. To address these constraints, we present the Child Detection Dataset (CDD), a dataset of 1,928 real-world images developed for child detection research. Unlike existing datasets, our proposed dataset captures children in more realistic, complex scenes. It includes images with multiple children, children interacting with adults, and children interacting with objects, thereby improving scene diversity and occlusion complexity. By incorporating images that reflect real-world scenarios, we aim to reduce the sim-to-real gap and improve the generalization capacity of child detection models. This contribution is intended to facilitate research on child detection, tracking, and action recognition, where robust performance is crucial for safety-critical applications such as surveillance, child monitoring, and early childhood education technologies.

Our contributions extend beyond technical advancement, offering direct clinical applications in video-based infant seizure detection and continuous monitoring. The methodology shows particular utility in home monitoring applications, where continuous professional observation proves impractical. Our framework establishes a foundation for automated pediatric assessment while maintaining robust privacy standards and ethical considerations. This work addresses critical gaps in current medical monitoring capabilities while opening new possibilities for automated diagnostic support in pediatric neurology.

The rest of the paper is organized as follows. Section II examines related work in child detection datasets and models, emphasizing medical applications. Section III details the dataset collection methodology and preprocessing procedures. Section IV presents experimental results and analysis. Section V describes implementation in child movement tracking in video-EEG. Section VI discusses implications and future research directions.

II. RELATED WORK

Over the past seven years, several datasets have been developed for child detection, tracking, and pose estimation [11], [12], [13]. These datasets have contributed to the development of computer vision models for understanding child behavior and movement. Current datasets exhibit specific limitations in scale, diversity, and representation of real-world scenarios [14].

The MIA dataset [15], introduced in 2017, consists of 35 sessions of 300-second depth video recordings of a single preterm infant at 37+1 weeks of gestational age. While the dataset is limited in size and subject variety, it established methods for data collection in clinical settings.

The Mini-RGBD dataset [16] introduced a combination of real and synthetic data, incorporating depth information alongside RGB data. This dataset enabled increased scale and diversity of poses while maintaining control over ground-truth annotations. However, the data set remained limited in the complexity of the scene and the variability of the real world. The BabyPose dataset [17] further improved data quality by focusing on consistency in data capture. It introduced recordings from 16 preterm infants, allowing for more comprehensive subject-level variability compared to single-infant datasets like MIA. BabyPose maintained controlled clinical capture conditions, similar to MIA, but benefited from a larger subject pool, allowing greater generalization in pose estimation tasks. However, the controlled environment of this dataset limited its applicability to real-world natural scenarios. The YouTube-Infant dataset [18], [19] shifted from controlled environments to natural video content, constructed from 85 videos of infants aged 0-6 months from public sources. This dataset introduced natural variability in the poses and environments of infants. However, the use of online platform data presented challenges in annotation accuracy and dataset consistency.

A more comprehensive approach to child pose estimation was presented with the introduction of the SyRIP dataset [9]. SyRIP is one of the largest child-related datasets to date, containing 1,700 images, of which 700 are real images and 1,000 are synthetic. This dataset was a significant step toward larger-scale hybrid datasets. By combining real-world images with synthetic data, SyRIP aimed to provide broader scene diversity. However, as with previous datasets, SyRIP's reliance on synthetic data introduces a "sim-to-real" gap, as the 3D-rendered synthetic images do not fully capture the visual complexity of real-world environments. Furthermore, the data set focuses largely on single-child images, lacking the complexity of multichild interactions, occlusions, or the presence of parents and caregivers. These elements are often encountered in real-world scenarios. In summary, while significant progress has been made in developing child-related datasets, several limitations remain [20], [21]. Most datasets are infant-centric, with few attempts to expand coverage to older children. Furthermore, scene complexity remains low, with many datasets capturing children in isolation, without the presence of parents, other children, or real-world clutter. The reliance on synthetic images is another limiting factor, as synthetic images lack the visual richness, variability, and occlusion complexities of real-world environments. Addressing these gaps is crucial for advancing child detection and pose estimation models, as more realistic datasets are required to train models capable of handling diverse, uncontrolled settings.

While the development of datasets has enabled progress in child detection research, the effectiveness of these datasets is intrinsically linked to the evolution of detection algorithms. Object detection, a core task in computer vision, has seen remarkable advancements in recent years due to the ongoing development of more efficient and accurate algorithms [22], [23]. One of the most significant breakthroughs in this field is the You Only Look Once (YOLO) framework [24]. YOLO's single-stage detection approach, which simultaneously predicts bounding boxes and class probabilities, has advanced through multiple iterations. However, few studies have specifically addressed child detection as a distinct computer vision task, with most frameworks, including YOLO variants, classifying children within the general 'person' category. This limitation becomes particularly apparent when tracking the movements of infants and children independently of adults. Recent studies demonstrate both the potential and limitations of current approaches. Lin et al. [25] introduced an anthropometric approach utilizing body proportions for age group classification, achieving detection accuracies of 95% and 92.5% for children and adults respectively, though limited to binary classification between subjects under 12 and over 15 years. Abdulghani et al. [26] employed the detection of YOLOv8m pre-trained persons for child safety applications, while Bisla et al. [27] focused on tracking a single child in a kindergarten setting using YOLOv8s. However, these studies highlight the persistent gap in addressing the unique challenges of general child detection, particularly in scenarios requiring precise tracking of child-specific movements and behaviors. This overview of datasets and detection methods reveals critical gaps in current approaches. Most datasets remain infant-centric with limited scene complexity, while detection methods lack specialization for child-specific characteristics. These limitations underscore the need for more comprehensive datasets and specialized detection approaches that can address the unique challenges of child detection in diverse, real-world settings.

III. THE NEW CHILD DETECTION DATASET—CDD

The dataset consists of videos obtained from Pexels.com, a stock video platform. The collection contains 358 videos of 331 different subjects documenting children from infancy through adolescence. All annotations in the dataset exclusively mark children as a single class (child), despite the presence of adults and other objects in videos. The selection process implemented specific criteria: each child must be identifiable by human observers, independent of their position as primary or secondary subjects in the frame. The videos contain varying numbers of children per scene, representing different age groups and physical characteristics. Videos were selected using specific search terms including 'children playing,' 'infants,' 'infant movement,' 'toddler activities' and 'teenagers' on Pexels.com. To ensure diversity, we selected samples across age groups, environments (indoor/outdoor), and interaction types. Each video underwent initial quality screening against inclusion criteria: (1) clearly visible children, (2) adequate lighting conditions, and (3) absence of severe motion blur.

Scene characteristics exhibit significant variation across the dataset. Age distribution analysis indicates baby subjects (0-12 months), toddlers (1-3 years), preschooler (3-5 years), child (5-11 years), and teenager (11-14). The number of subjects per frame ranges from 1 to 8 children but we can also find adults in the same videos. Subjects are likely to be occluded by an object or other subject / adults. Environmental contexts include indoor settings, outdoor environments, and structured environments such as classrooms or play areas. Lighting conditions vary between natural daylight, artificial indoor lighting, and mixed lighting conditions. Subjects can be found interacting in complex manner with other subject or object. The dataset was randomly split into training (57.16%) and testing (42.84%) sets. We enforced strict video-level separation, ensuring no overlapping clips between sets, while permitting subject-level overlap when individuals appeared in different videos. The frame extraction protocol established a maximum of 10 keyframes per video sequence, with complete frame extraction implemented for videos containing fewer than 10 keyframes. Each extracted frame underwent standardization to a 640 × 640 pixel resolution, maintaining compatibility with our fine-tuned infant detection model. Each frame was manually reviewed to confirm quality and subject visibility before inclusion in the final dataset. The annotation process followed a standardized protocol focusing exclusively on child detection. While videos may contain adults and other objects, annotators labeled only children (single class) using bounding boxes. The annotation process followed a standardized protocol and was conducted by the first and last (fourth) authors of the manuscript using the Computer Vision Annotation Tool (CVAT). Annotators were trained with a standardized guideline that included visual examples illustrating correct bounding box placement for children across a range of poses and occlusion conditions. The resulting annotations were validated by the third author, who serves as the head of the pediatric intensive care unit at Raymond Poincaré Hospital.

The methodological pipeline for dataset construction is illustrated in Figure 2, with quantitative characteristics of the final dataset composition detailed in Table 1. The dataset can be found at https://www.kaggle.com/datasets/samueldiop/child-detection-dataset

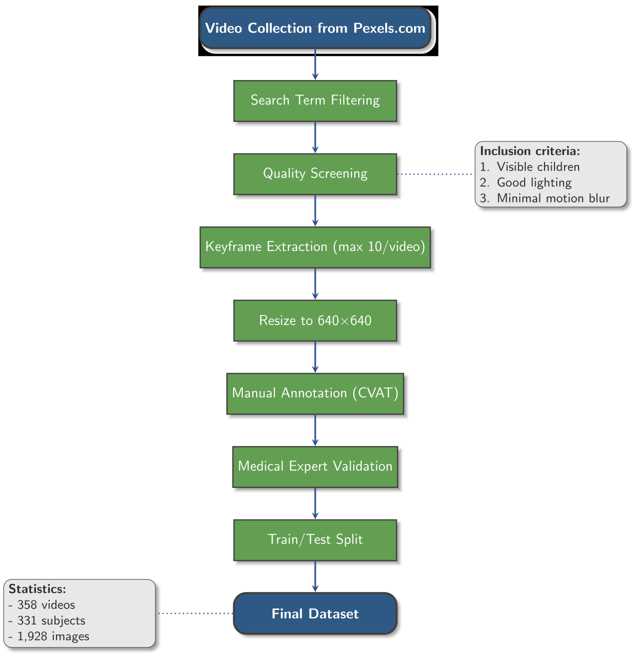


FIGURE 2. CDD dataset creation pipeline showing the systematic process from video collection to final dataset preparation, including quality control measures and integration with the SyRIP dataset.

Table 2 provides a systematic comparison of six child development datasets, highlighting key distinctions in scale, data modality, and application context. Our CDD dataset (N=331) offers the most comprehensive coverage (0–14 years) with real-world diversity, while specialized datasets like MIA (N=1, depth videos) and BabyPose (N=16, preterm limb analysis) focus on controlled medical scenarios. Notably, SyRIP (N=140) and MINI-RGBD (N=12) provide synthetic data alternatives, whereas

TABLE 1. CDD dataset statistics, including age distribution, occlusion levels, and interaction types. Occlusion subtypes define visibility thresholds (*Partial*: >50% body visible; *Severe*: <50% visible).

Category	Type	Subtype	Count
Age Group	0–1 years	—	46
	1–3 years	—	47
	3–5 years	—	38
	5–11 years	—	178
	11–14 years	—	22
Occlusion Level	None	>50% visible	142
	Partial	>50% visible	148
	Severe	<50% visible	68
Interaction Type	Single-person	—	50
	Two-person	—	87
	Group	—	116
	Object-based	—	105

YouTube-Infant (N=104) captures naturalistic behavior without clinical annotations. This comparison underscores CDD's unique combination of scale, realism, and multi-child interactions—addressing gaps in existing resources.

IV. *YOLO*$_{\mathbf{CDD}}$: A FINE-TUNED MODEL FOR CHILD DETECTION

YOLO represents a significant contribution in object detection, distinguished by its single-stage architecture that simultaneously predicts bounding boxes and class probabilities in one forward pass. Among the YOLO family, YOLOv11 stands as the latest iteration, bringing substantial improvements in detection accuracy and computational efficiency [28]. This study presents YOLO-Child Detection Dataset, YOLOCDD, a model adapted from YOLOv11x specifically for child detection tasks.

The model architecture maintains YOLOv11x's fundamental three-component structure while modifying the detection parameters for child-specific features. The backbone component performs initial feature extraction through CNN layers and C3k2 blocks, implementing an optimized Cross Stage Partial Bottleneck design. A Spatial Pyramid Fast module processes features at multiple scales, enabling adaptation to varying input dimensions. The model's neck component integrates features across different scales through a sequence of convolutional layers, upsampling operations, and feature concatenation. C3k2 blocks process these combined features before transmission to the detection head. The head component, modified for single-class detection, generates the final outputs through C3k2 blocks, followed by Convolution-BatchNorm-SiLU (CBS) layers. A final 2D convolution produces bounding box coordinates and objectness scores specifically calibrated for child detection. Figure 3 illustrates our fine-tuning pipeline, where the backbone and neck components remain frozen while the detection head is optimized for child-specific features.

TABLE 2. Comparison of Child Development Datasets (CDD, SyRIP, MIA, BabyPose, MINI-RGBD, and YouTube-Infant).

Dataset	Subjects	Data Type	Age Range	Real-World Complexity	Multi-Child Interactions	Medical Scenarios
CDD	331	Real images	0-14 years	Extensive and diverse	Comprehensive	Some medical cases
SyRIP	140	Synthetic/Real Images (2D joints)	0-12 months	Moderate (specific scenarios)	Limited/absent	Minimal
MIA	1	Depth videos	37±1 weeks	Moderate (controlled)	Not applicable	NICU preterm monitoring
BabyPose	16	Depth videos (limb-joints)	24-37 weeks	Lower complexity	Not applicable	Preterm movement analysis
MINI-RGBD	12	Synthetic videos (RGB+D, 2D-3D joints)	0-7 months	Controlled (lab setting)	Not applicable	Developmental studies
YouTube-Infant	104	Videos (BINS score)	6-9 weeks	High diversity	Not applicable	Non-clinical settings

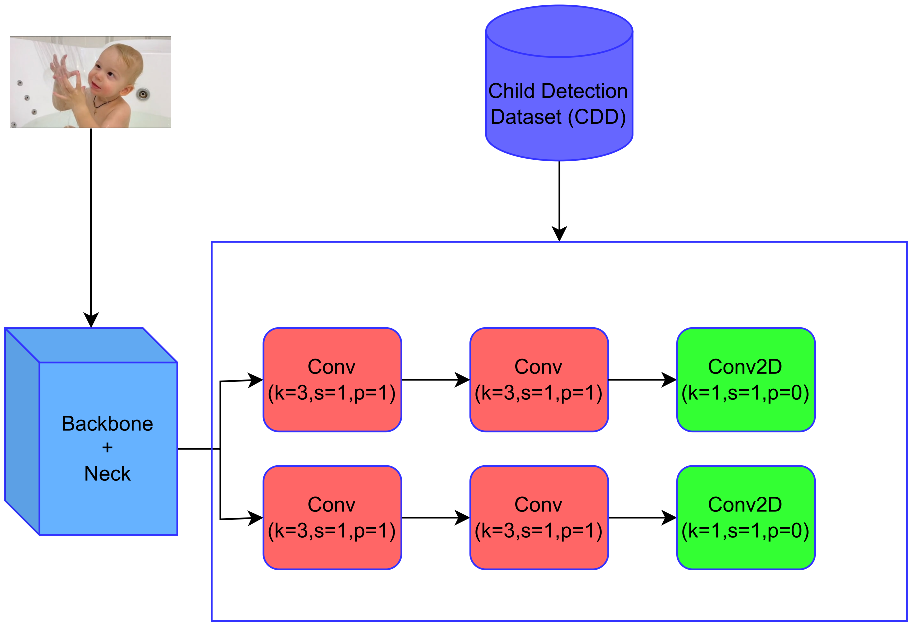


FIGURE 3. Training of YOLOCDD. The Backbone and head of YOLO were frozen and the head was fine-tuned on the Child Detection Dataset (CDD).

Our implementation utilizes transfer learning by freezing the first 20 layers of YOLOv11x, preserving the fundamental feature extraction capabilities while allowing specialization in the terminal neck and head components. This architectural decision maintains the network's ability to detect basic visual elements while enabling targeted adaptation for child-specific features. The model processes standardized input images of 640 × 640 pixels, maintaining architectural compatibility while ensuring consistent detection performance. Each frame undergoes analysis through a 64 × 64 matrix, enabling both localized detection and global scene understanding. The single-class adaptation simplifies the detection task while maintaining high precision in child identification. This specialized architecture demonstrates how transfer learning can be effectively applied to create task-specific detection models while preserving the computational efficiency of the YOLO framework. The model's focus on child-specific detection parameters enables precise localization across various scenarios, supporting applications requiring dedicated child tracking and analysis.

TABLE 3. Dataset distribution for model evaluation.

Dataset	Training Samples	Testing Samples	Total
SyRIP	1,087	113	1,200
CDD	1,102	826	1,928

V. RESULTS

A. COMPARATIVE ANALYSIS OF DIFFERENT YOLO VARIANTS

We evaluate our YOLO models on two datasets: (a) our Child Detection Dataset CDD and (b) the SyRIP dataset, selected as the largest comparable public benchmark. This dual evaluation tests performance across both controlled (SyRIP) and diverse real-world (CDD) environments.

Table 3 summarizes the distribution of samples across training and testing sets.

For all experiments, we tuned several key hyperparameters: learning rate (lr) ranging from $5.0e^{-6}$ to 0.01, fractional learning rate (flr) from 0.10 to 0.50, number of epochs (30 or 150), and loss function components including box detection importance (7.50-10.00).We shared the top 2 results (except table 10 that contains only top 1 results) on the testing split on the dataset study.

The training was conducted on a NVIDIA RTX 3090 graphic card with 24 go of VRAM and a AMD Ryzen 9 5000 Series cpu, with a typical training run requiring approximately 1 hours for 30 epochs on our combined dataset.

We conducted comprehensive testing using three variants of the YOLO11 architecture: YOLO11x, which offers enhanced feature extraction through its expanded network capacity, YOLO11m, which provides a balanced trade-off between computational efficiency and detection accuracy; and YOLO11n, a lightweight variant optimized for resource-constrained environments while maintaining acceptable detection performance. For all experiments, we calculated precision, recall, and Mean Average Precision (mAP) metrics following the COCO evaluation protocol. mAP50 represents the average precision at 50% IoU threshold, while mAP50-95 averages precision across IoU thresholds from 0.5 to 0.95 in 0.05 increments, providing insight into both detection capability and localization accuracy.

For models without a dedicated 'child' class (e.g., YOLOv11 and its variants), evaluation is performed using the predicted bounding box closest to the ground-truth child annotation, ensuring alignment with the task's focus.

Tables 4 and 5 present our detailed experimental results across different dataset configurations. Notably, YOLOv11x demonstrated superior performance in most scenarios, particularly when trained for 150 epochs on the SyRIP dataset, achieving remarkable mAP@50 of 0.995 and mAP@50-95 of 0.885. Figure 4 illustrates the model's detection capability on test set examples.

B. MEDICAL CASE STUDY: CHILD MOVEMENT TRACKING IN VIDEO-EEG

We assessed the efficacy of YOLOCDD for child detection in video-EEG examinations, where accurate and consistent tracking is medically essential. The study focused on uncontrolled clinical environments characterized by complex interactions among children, parents, and medical staff. Such settings introduce significant variability, including occlusions from EEG helmets, dynamic pose changes, and overlapping individuals, presenting substantial challenges for robust object detection. The evaluation dataset consisted of seven annotated video sequences capturing real-world video-EEG sessions. These sequences depicted pediatric patients interacting with caregivers and clinicians while wearing EEG headgear, resulting in frequent occlusions and non-standard poses. Two independent annotators, both board-certified neurologists, established the ground truth labels for child detection. To benchmark performance, we compared YOLOCDD against YOLOv11x, with detailed results presented in Table 6. The evaluation emphasized localization precision by measuring the pixel-wise displacement between the predicted person detections from YOLOv11x and the ground truth child coordinates. This approach ensured a clinically relevant assessment of detection accuracy in crowded, occlusion-prone scenarios typical of medical monitoring. Our comparative evaluation revealed YOLOCDD 's superior performance in child detection tasks within clinical environments. As demonstrated in Figure 5, YOLOCDD maintained robust detection accuracy even in challenging scenarios where the child and attending physician appeared similar in size due to the physician's seated position. Notably, the model correctly discriminated the child as the target while rejecting false positives from medical staff, showcasing its domain-specific optimization. Quantitative analysis showed YOLOCDD achieving a precision of 0.966 and mAP@50 of 0.953, representing significant improvements over YOLOv11x's respective scores of 0.595 and 0.606. While YOLOv11x exhibited a marginally higher mAP@50-95 (0.501 versus 0.423), this advantage likely stems from its COCO dataset pre-training and architectural emphasis on localization precision rather than clinical specificity. The clinical relevance of YOLOCDD 's performance becomes particularly evident when examining tracking stability metrics. With a movement tracking success rate of 85.7% compared to YOLOv11x's 84.2%, coupled with its substantially better precision, YOLOCDD proves more effective for continuous medical monitoring where consistent detection outweighs the need for exact boundary delineation.

VI. DISCUSSION

Our experimental results establish that YOLOCDD significantly outperforms general-purpose detection models in child-specific recognition tasks, achieving a precision of 0.966 compared to YOLOv11x's 0.595. This performance advantage stems from three key architectural innovations: (a) strategic layer freezing that preserves fundamental feature extraction while enabling child-specific adaptation, (b) optimized terminal layers for pediatric detection, and (c) domain-specific training data curation. The model's 85.7% tracking success rate in video-EEG monitoring demonstrates its clinical viability, particularly in overcoming challenges posed by medical equipment and staff interactions. The transfer learning approach implemented through partial layer freezing proved particularly effective, maintaining the model's ability to process complex medical environments while specializing its higher-level features for child detection. This design choice effectively addresses a core challenge in medical computer vision: balancing general visual understanding with domain-specific precision. Our results demonstrate that the frozen layers robustly manage variability in lighting conditions and equipment occlusion, while the trainable layers adapt to child-specific visual features essential for diagnostic purposes. However, several limitations warrant consideration. The current dataset, though carefully curated for medical scenarios, remains limited in scale compared to general object detection benchmarks. This relative scarcity of training data may impact the model's ability to generalize to rare but

TABLE 4. Performance evaluation on SyRIP dataset. "flr" and "box" represent focal loss ratio and box loss weight respectively. Bold values indicate best performance.

Model	Epochs	lr	Key Hyperparameters		Performance	
			flr	box	mAP_{50}	$mAP_{50\text{-}95}$
YOLO11x	150	5.0×10^{-6}	0.10	9.00	**0.995**	**0.885**
YOLO11x	30	5.0×10^{-6}	0.10	9.00	0.995	0.871
YOLO11m	30	1.0×10^{-2}	0.50	7.50	0.995	0.876
YOLO11n	30	5.0×10^{-6}	0.50	9.00	0.995	0.846

TABLE 5. Performance evaluation on CDD dataset."flr" and "box" represent focal loss ratio and box loss weight respectively. Bold values indicate best performance.

Model	Epochs	lr	Hyperparameters		Performance	
			flr	box	mAP_{50}	$mAP_{50\text{-}95}$
YOLO11x (150)	150	5.0×10^{-6}	0.10	9.00	**0.828**	**0.477**
YOLO11x (30)	30	5.0×10^{-6}	0.10	9.00	0.795	0.406
YOLO11m	30	1.0×10^{-2}	0.50	7.50	0.841	0.486
YOLO11n	30	5.0×10^{-6}	0.50	9.00	0.765	0.407

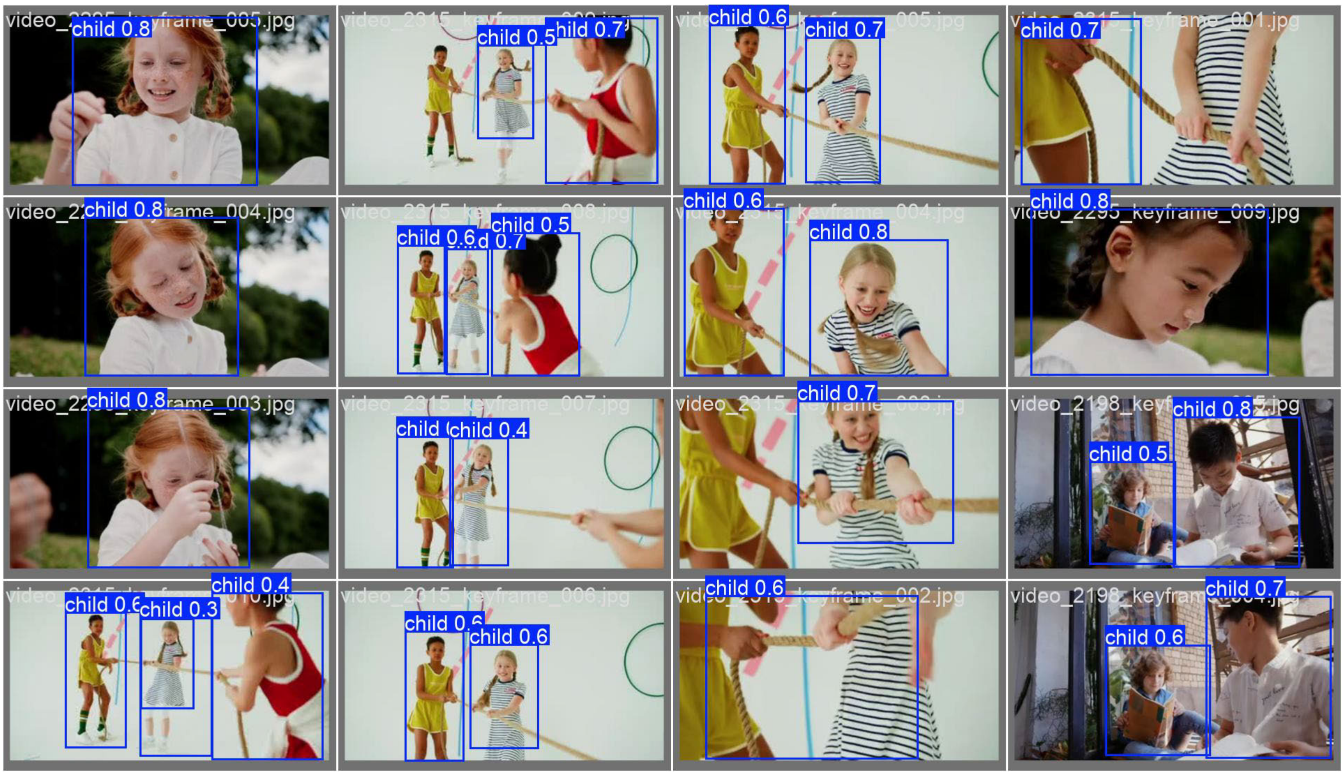
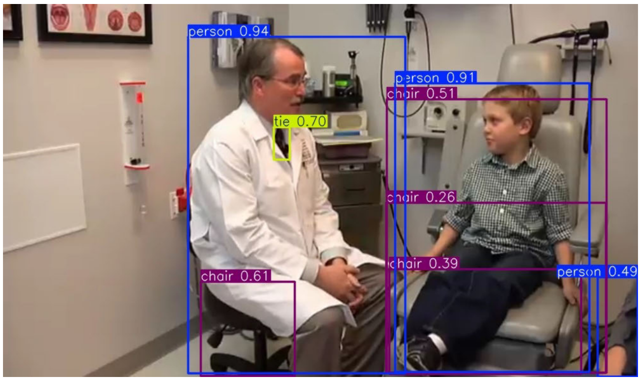
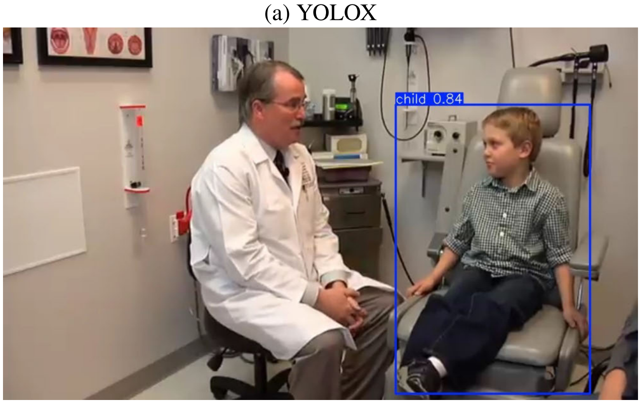


FIGURE 4. Detection results of the YOLOCDD model on the test set, demonstrating its performance across diverse scenarios, including indoor and outdoor environments, as well as varying levels of occlusion.

clinically important cases, such as children in unconventional poses or under unusual lighting conditions common in hospital environments. Furthermore, the exclusive focus on child labels, while beneficial for targeted pediatric applications, currently prevents the model from distinguishing between children and adults in mixed populations. The medical case studies reveal another important consideration: while YOLOCDD excels at consistent detection, the slight compromise in boundary precision (as reflected in mAP50-95 metrics) may require attention for certain diagnostic applications where precise spatial localization proves critical. This trade-off between detection reliability and localization accuracy represents a fundamental challenge in medical computer vision that warrants further investigation.



(a) YOLOX



(b) YOLOCDD

FIGURE 5. Comparative detection results of YOLOX (at the top) and YOLOCDD (at the bottom) models on EEG-Video exams conducted at Raymond Poincaré Hospital (APHP) in Garches. The public images were chosen because they closely resemble our medical dataset, which cannot be shared due to legal restrictions. The images illustrate the detection of children in a clinical environment, including interactions with medical staff and the presence of EEG equipment.

TABLE 6. Comparison between YOLOCDD and YOLOv11x on medical scenes.

model	Precision	Recall	mAP50	mAP50-95
YOLOCDD	**0.966**	**0.857**	**0.953**	0.423
YOLOv11x	0.595	0.842	0.606	0.501

Additionally, the current validation, though promising, has been conducted in relatively controlled clinical environments; broader testing across diverse medical facilities with varying equipment and protocols would strengthen confidence in the model's robustness.

VII. CONCLUSION

This work makes three principal contributions to child detection research and medical computer vision applications. First, we introduce the CDD dataset, a curated collection of 1,928 real-world images capturing diverse child interactions, with particular emphasis on challenging medical environments. This resource directly addresses the critical shortage of specialized datasets for pediatric monitoring applications, enabling development of more robust detection systems. Second, we present YOLOCDD, a novel architecture demonstrating significant improvements in detection accuracy (precision: 0.966, mAP@50: 0.953) over general-purpose alternatives. Third, we validate our approach in clinically-relevant video-EEG monitoring scenarios, where the model maintains reliable performance despite common obstructions from medical equipment and staff interactions. Several promising directions emerge for future research. One of the most immediate opportunities lies in the expansion of the dataset to include more diverse medical scenarios and equipment configurations would further enhance model robustness. In parallel, extending the model's detection capabilities beyond children to include adults could significantly benefit multi-person monitoring contexts. By integrating these annotations, the system could evolve into a multi-class detector, capable of distinguishing between children, caregivers, and healthcare professionals. Another promising avenue involves leveraging action recognition models that have shown success in adult populations. For example, the I3D neural network, Originally designed for action recognition in single subjects, and later applied epilepsy detection for adults, could be adapted and extended to address the specific challenges of epilepsy detection in children, offering a tailored solution that aligns with the unique behavioral and physiological patterns observed in pediatric patients. By leveraging such specialized models, we can further enhance the effectiveness and precision of child-focused medical monitoring systems, paving the way for significant advancements in this critical area. Together, these directions point toward a future in which child detection systems are not only more accurate, but also more intelligent and context-aware.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The collection, transfer, and storage of the medical dataset adhered to stringent compliance with the General Data Protection Regulation GDPR and the CNIL. Additionally, the Ethical Scientific Committee for research, studies, and evaluation in the health domain (CESREES) has approved the utilization of this data for our research, under protocol number 20240119163135. All identifiable patient information in the case study was removed from the video data obtained prior to processing. Informed consent was obtained from each study participant's caregivers or from capable patients aged over 18 years for the publication of their data.

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REFERENCES

[1] E. Keles and U. Bagci, "The past, current, and future of neonatal intensive care units with artificial intelligence: A systematic review," *NPJ Digit. Med.*, vol. 6, no. 1, p. 220, Nov. 2023.

[2] S. Diop, N. Essid, F. Jouen, J. Bergounioux, and I. Trabelsi, "Adapting action recognition neural networks for automated infantile spasm detection," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 3751–3760, 2024.

[3] M. Javaid, A. Haleem, R. P. Singh, and M. Ahmed, "Computer vision to enhance healthcare domain: An overview of features, implementation, and opportunities," *Intell. Pharmacy*, vol. 2, no. 6, pp. 792–803, Dec. 2024.

[4] P. Rai, A. Knight, M. Hiillos, C. Kertész, E. M. Salinas, D. Terney, S. A. Larsen, T. Østerkjerhuus, J. Peltola, and S. Beniczky, "Automated analysis and detection of epileptic seizures in video recordings using artificial intelligence," *Frontiers Neuroinform.*, vol. 18, Mar. 2024, Art. no. 1324981.

[5] D. F. Huelke, "An overview of anatomical considerations of infants and children in the adult world of automobile safety design," in *Proc. Annu. Assoc. Advancement Automot. Med.*, vol. 42, Jan. 1998, pp. 93–113.

[6] E. Sedgh-Gooya, F. Aram, W. Kaddah, M. Elbouz, and A. Alfalou, "A human body morphology detector: A distinctive filter to differentiate the image-based process depending on whether the person is a child or an adult," *Proc. SPIE*, vol. 11735, pp. 176–180, Jul. 2021.

[7] R. Falsaperla, S. Sciuto, G. F. Privitera, L. G. Tardino, G. Costanza, A. Di Nora, R. H. Caraballo, and M. Ruggieri, "Epileptic spasms in infants: Can video-EEG reveal the disease's etiology? A retrospective study and literature review," *Frontiers Neurol.*, vol. 14, Jun. 2023, Art. no. 1204844.

[8] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.

[9] X. Huang, N. Fu, S. Liu, and S. Ostadabbas, "Invariant representation learning for infant pose estimation with small data," in *Proc. 16th IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, Dec. 2021, pp. 1–8.

[10] I. Milkaite and E. Lievens, "Children's rights to privacy and data protection around the world: Challenges in the digital realm," *Eur. J. Law Technol.*, vol. 10, no. 1, pp. 1–24, 2019.

[11] M. M. E. Yurtsever and S. Eken, "BabyPose: Real-time decoding of baby's non-verbal communication using 2D video-based pose estimation," *IEEE Sensors J.*, vol. 22, no. 14, pp. 13776–13784, Jul. 2022.

[12] G. Sciortino, G. M. Farinella, S. Battiato, M. Leo, and C. Distante, "On the estimation of children's poses," in *Proc. Int. Conf. Image Anal. Process.*, 2017, pp. 410–421.

[13] Q. Wu, G. Xu, F. Wei, J. Kuang, P. Qin, Z. Li, and S. Zhang, "Supine infant pose estimation via single depth image," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[14] T. Olugbade, M. Bieńkiewicz, G. Barbareschi, V. D'amato, L. Oneto, A. Camurri, C. Holloway, M. Björkman, P. Keller, M. Clayton, A. C. D. C. Williams, N. Gold, C. Becchio, B. Bardy, and N. Bianchi-Berthouze, "Human movement datasets: An interdisciplinary scoping review," *ACM Comput. Surv.*, vol. 55, no. 6, pp. 1–29, Jun. 2023.

[15] A. Cenci, D. Liciotti, E. Frontoni, P. Zingaretti, and V. P. Carnielli, "Movements analysis of preterm infants by using depth sensor," in *Proc. 1st Int. Conf. Internet Things Mach. Learn.*, Oct. 2017, pp. 1–9.

[16] K. D. McCay, E. S. L. Ho, C. Marcroft, and N. D. Embleton, "Establishing pose based features using histograms for the detection of abnormal infant movements," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2019, pp. 5469–5472.

[17] L. Migliorelli, S. Moccia, R. Pietrini, V. P. Carnielli, and E. Frontoni, "The babyPose dataset," *Data Brief*, vol. 33, Dec. 2020, Art. no. 106329.

[18] C. Chambers, N. Seethapathi, R. Saluja, H. Loeb, S. R. Pierce, D. K. Bogen, L. Prosser, M. J. Johnson, and K. P. Kording, "Computer vision to automatically assess infant neuromotor risk," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 11, pp. 2431–2442, Nov. 2020.

[19] H. Ni, Y. Xue, L. Ma, Q. Zhang, X. Li, and X. Huang, "Semi-supervised body parsing and pose estimation for enhancing infant general movement assessment," *Med. Image Anal.*, vol. 83, Oct. 2022, Art. no. 102654.

[20] M. Leo, G. M. Bernava, P. Carcagnì, and C. Distante, "Video-based automatic baby motion analysis for early neurological disorder diagnosis: State of the art and future directions," *Sensors*, vol. 22, no. 3, p. 866, Jan. 2022.

[21] N. Silva, D. Zhang, T. Kulvicius, A. Gail, C. Barreiros, S. Lindstaedt, M. Kraft, S. Bölte, L. Poustka, K. Nielsen-Saines, F. Wörgötter, C. Einspieler, and P. B. Marschik, "The future of general movement assessment: The role of computer vision and machine learning—A scoping review," *Res. Develop. Disabilities*, vol. 110, Mar. 2021, Art. no. 103854.

[22] Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," *Proc. IEEE*, vol. 111, no. 3, pp. 257–276, Mar. 2023.

[23] A. K. Gupta, A. Seal, M. Prasad, and P. Khanna, "Salient object detection techniques in computer vision—A survey," *Entropy*, vol. 22, no. 10, p. 1174, Oct. 2020.

[24] P. Jiang, D. Ergu, F. Liu, Y. Cai, and B. Ma, "A review of YOLO algorithm developments," *Proc. Comput. Sci.*, vol. 199, pp. 1066–1073, Jan. 2022.

[25] J.-M. Lin, W.-L. Lin, and C.-P. Fan, "Age group classifier of adults and children with YOLO-based deep learning pre-processing scheme for embedded platforms," in *Proc. IEEE 12th Int. Conf. Consum. Electron. (ICCE-Berlin)*, Sep. 2022, pp. 1–3.

[26] A. M. Abdulghani, M. M. Abdulghani, W. L. Walters, and K. H. Abed, "Real-time object detection to identify adults and children using YOLO algorithms," in *Proc. Int. Conf. Comput. Sci. Comput. Intell. (CSCI)*, Dec. 2023, pp. 1146–1151.

[27] T. Bisla, R. Shukla, M. Dhawan, M. R. Islam, T. Koyasu, K. Teramoto, Y. Kataoka, and K. Horio, "Kid activity recognition: A comprehensive study of kid activity recognition with monitoring activity level using YOLOv8s algorithms," in *Proc. 3rd Int. Conf. Artif. Intell. Internet Things (AIIoT)*, May 2024, pp. 1–6.

[28] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.

SAMUEL DIOP received the bachelor's degree in computer science from Paris-Saclay University, Orsay, France, and the master's degree from California State University San Marcos, San Marcos, CA, USA. He is currently pursuing the Ph.D. degree with the École Pratique des Hautes Études (EPHE/PSL), Paris, France. He completed his Informatics Technology Expert Certification at Epitech, Paris. He is a member of the CHArt Laboratory, where his research focuses on machine learning-driven detection of infantile epilepsy, particularly infantile spasms. Previously, he was a Research Engineer at EPHE, a Research Intern at Inserm, and a Backend Developer Intern at Fealing. His technical expertise spans computer vision, Bayesian statistics, and open-source tools within the Python ecosystem. He received the Outstanding Student Award, in 2020.

FRANÇOIS JOUEN received the Ph.D. degree from Université Paris V, Paris, France, and the H.D.R. degree in life and health sciences from Université Paris VI, Paris. His Ph.D. thesis focused on the development of visuo-vestibular interactions in infants. He held a postdoctoral fellowship at Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, where he designed psychophysical experiments for orbital flight studies. He is currently a Professor Emeritus with the École Pratique des Hautes Études, Paris. Since 2021, he has been leading the R2P2 platform, Raymond Poincaré Hospital. Previously, he co-directed the CHArt Research Unit and was a Researcher at French National Center for Scientific Research. His work bridges developmental neuroscience and technology, emphasizing epigenetic approaches to human development, neurodevelopmental risk prevention in infants, and the creation of low-cost tools for physiological monitoring. He pioneered infrared thermography for pain measurement in preterm neonates, video analysis of spontaneous infant motility, and the redCV computer vision library for automated movement tracking.

JEAN BERGOUNIOUX received the joint master's degree in human genetics from Université Paris VII, Paris, France, and King's College London, London, U.K., in 2001, the M.D. degree from Université Paris XI, Paris, in 2004, the Pediatric Specialist degree from Université Paris XI, in 2005, and the joint Ph.D. degree in cellular and molecular microbiology from Université Paris VII and the Pasteur Institute, Paris, in 2009. He has been a Full Professor of pediatrics with the UFR Simone Veil, Infection and Inflammation Department, Université Versailles Saint-Quentin-Paris Saclay, France, since 2020. Since 2021, he has been the Head of the Pediatric Neurology and Intensive Care Department, Raymond Poincaré Hospital (APHP). His work integrates clinical expertise in severe pediatric diseases with innovative research on AI-driven diagnostic systems, 3D-printed medical devices, and automated monitoring tools for vulnerable patients. He has led several research projects funded by Microsoft Azure, Aerogen, APHP, and the AXA Foundation.

IMEN TRABELSI received the Ph.D. degree in signal processing and machine learning from the National School of Engineers of Tunis (ENIT), Tunisia. Following her doctoral studies, she relocated to France, where she pursued research in artificial intelligence applied to healthcare and multimodal data analysis. She is currently a Research Associate with the Research Unit (R2P2), Raymond Poincaré Hospital, Garches, France. Previously, she was a Research Fellow with French National Center for Scientific Research and a Researcher with INSA Rouen, Rouen, France. Her work focuses on developing AI-driven solutions for processing heterogeneous data, including video, time-series, and audio signals, to support clinical decision-making in pediatric neurology and critical care. She emphasizes end-to-end system design, from raw data acquisition to generating interpretable results for diverse audiences. She has authored multiple peer-reviewed papers in international conferences and journals. Her research interests span multimodal AI integration, automated signal analysis, and ethical AI frameworks for healthcare.